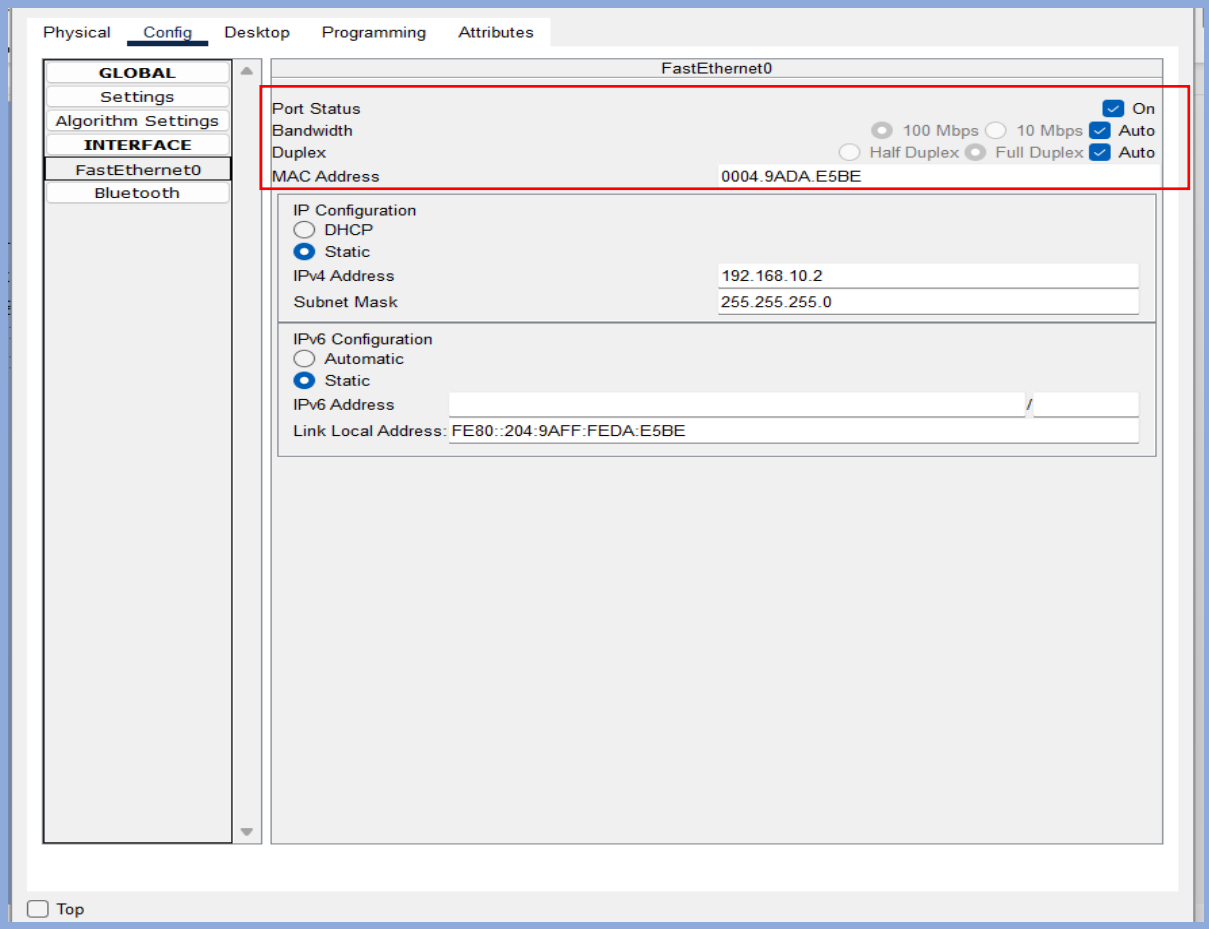
CTECT 22061
System and Network Laboratory

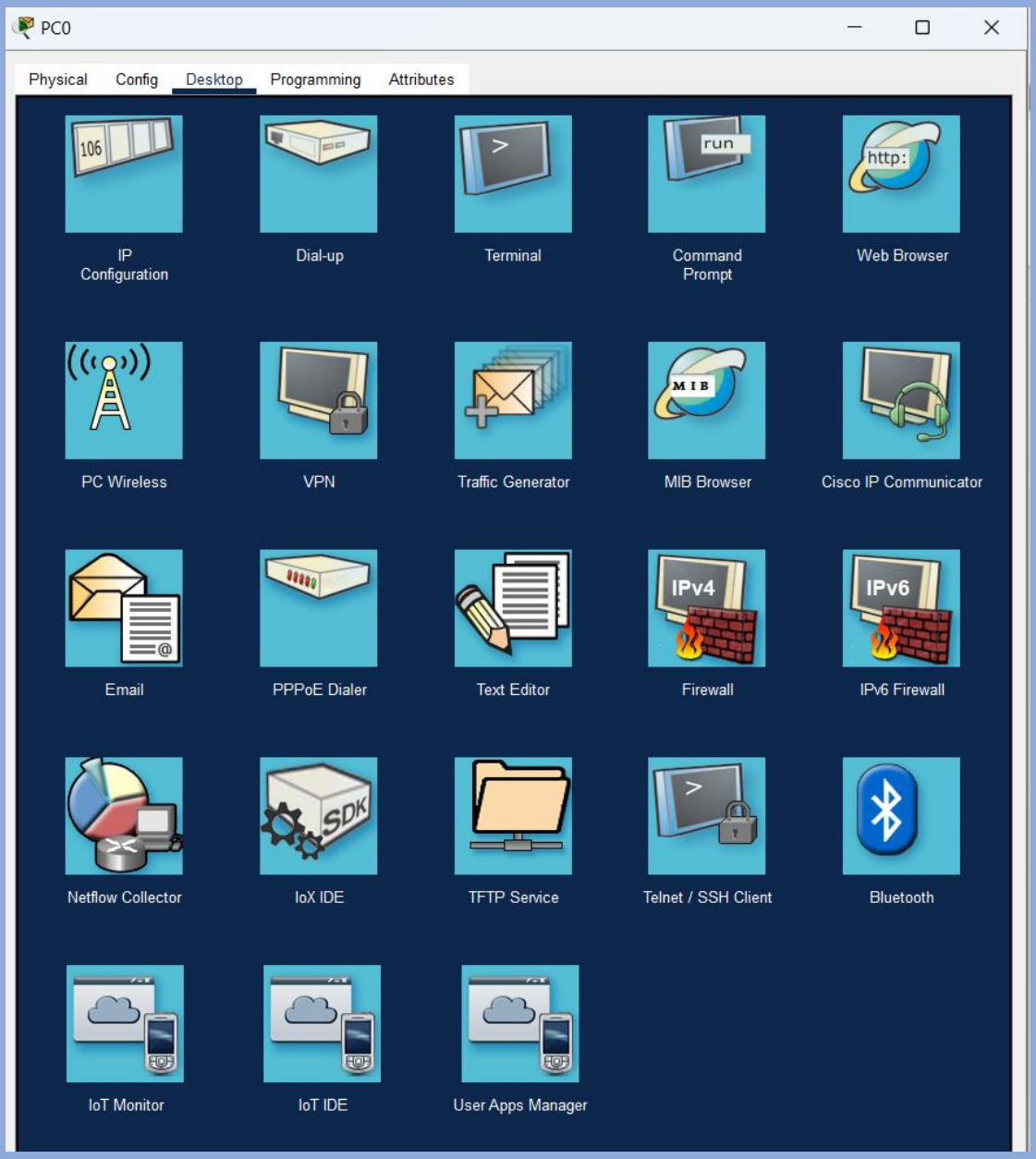
Day 2 Lab – implementing LANs

1. Connect two PCs with a switch using suitable cables.
2. Click on PC. Navigate to **Config** tab. Select "Static" option.
3. Assign IP addresses as follows and Subnet mask.
   192.168.10.2
   192.168.10.3

   What is the MAC address on your device? Attach a screenshot



4. Navigate to the **Desktop** tab. Check and confirm the configuration using “IP Configuration” “Command Prompt” sections. Search and learn about other sections in **Desktop** tab.



5. Change the IP configuration option as DHCP. What are the IPs that your PCs have? Attach a screenshot

6. Connect a server to the same topology. Let's configure the server as a DHCP server. Click on the server icon on the Packet Tracer Logical workspace and select the **Services** tab. Select DHCP from the SERVICES list in the left pane.

7. In the DHCP configuration window, configure a DHCP as shown in the figure with the following settings.
   - Click On to turn the DCHP service on
   - Pool name: DHCPpool
   - Starting IP Address: 208.67.220.1
   - Subnet Mask 255.255.255.0
   - Maximum number of Users: 50

   Click Add to add the DHCP service settings.

8. Configure the server FastEthernet0 Interface settings. Click on FastEthernet in left pane of the Config tab. Configure the FastEthernet Interface settings of the server as follows:
   - Select Static under IP Configuration
   - IP Address: 208.67.220.220
   - Subnet Mask: 255.255.255.0

9. Check the IP of your PCs. Is it from DHCP server IP range? Attach a screenshot

10. Connect another server to the same topology. Let's configure the server as a DNS server to provide domain name to IPv4 address resolution. Still in the Services tab, select DNS from the SERVICES listed in the left pane. Configure the DNS service using the following settings as shown in the figure.
    - Click On to turn the DNS service on
    - Name: Cisco.com
    - Type: A Record
    - Address: 208.67.220.220

    Click Add to add the DNS service settings.

11. Add the DNS address to server global settings.

12. Verify that the PC is receiving IPv4 configuration information and DNS from DHCP server. Click on the Command Prompt icon. In the command prompt refresh the IP settings by issuing the commands **ipconfig /release** and then **ipconfig /renew**. Attach a screenshot

13. Use **Ping** command to check the connectivity between PCs and servers.

    Attach a screenshot

14. Connect more PCs to the switch and configure them as above learned steps. How does a switch decide the path of a frame it receives from a connected device? Write the answer.

15. Click the switch and Open **CLI** tab on the switch. Set the basic configurations to the Switch as follows.

```
Switch>enable
Switch# configure terminal

Switch(config)# hostname SW-01
SW-01# config terminal
SW-01(config)# enable password CTEC22061
SW-01(config)# enable secret CTEC22061
SW-01(config)#banner motd # Unauthorised access is strictly prohibited #

SW-01(config)# line console 0
SW-01(config-line)# password CTEC22061
SW-01(config-line)# login
SW-01(config-line)# exit

Switch(config)#line vty 0 15
Switch(config-line)#password fctlabpassline
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#exit
```

16. Save the configuration

```
SW-01# copy running-config startup-config
SW-01# write

SW-01# show  running-config
SW-01# show startup-config
```

17. View and change the MAC address table.

```
SW-01# sh mac address-table
SW-01# clear mac address-table dynamic
SW-01# clear mac address-table dynamic address given MAC address
SW-01# clear mac address-table dynamic interface  Interface-ID
```

What data can be seen in your MAC table? Attach a screenshot

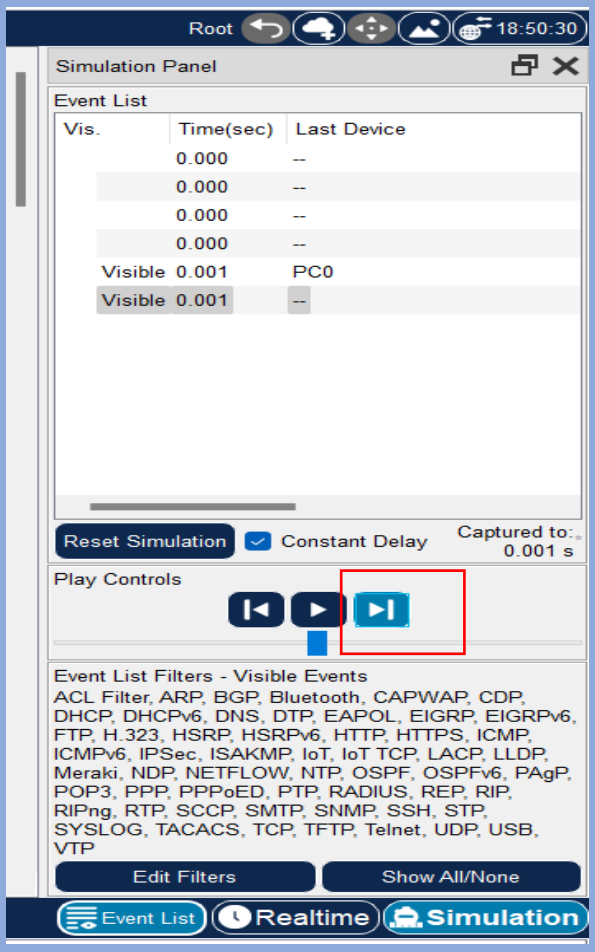
18. Let’s learn about Simulation mode

Add a simple PDU from common tool bar. Click on one PC and then on another PC. Then navigate to **Simulation mode**

Click Show **All/None tab** to filter the packets based on using protocols. Here select ICPM checkbox.

Use **capture and forward button** to investigate the packet path.

Click on a packet in **Simulation panel.** Investigate **inbound/ outbound PDU details.**



Now again check the MAC address on Switch. Attach a screenshot